Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. How can you limit the total permitted vocabulary size of a vectorizer to a specific value for performance reasons? 1 / 1 point

- Increase the ngram_range
- Set the min_df parameter
- Set the max_features property

Nice work

The 'max_features' property can be set to limit the number of terms included in the vocabulary based on their frequency, thus managing the total vocabulary size for performance optimization.

2. When implementing a bag-of-words model using scikit-learn, which class should be used to convert a list of document strings into a numeric term-document matrix? 1 / 1 point

- LabelEncoder
- OneHotEncoder
- TextMatrixCreator
- CountVectorizer

Nice work

The CountVectorizer class in scikit-learn is used to transform a list of document strings into a sparse numeric array where each cell contains the count of how many times each term occurs in each document.

3. True or False: The text processing parameter settings used to create a text vectorizer don't have a significant effect on the accuracy of downstream text classification tasks. 1 / 1 point

- True
- False

Nice work

Adjusting parameters in the vectorizer class alters how text is processed, which can significantly impact the overall quality and accuracy of the text processing pipeline.

4. A data scientist is working on a text analysis project involving a large collection of documents. They notice that terms like 'the', 'of', and 'but' are very frequent. What vectorizer parameter setting should they consider using to improve the efficiency of their text processing? 1 / 1 point

- stopwords
- ngram_range
- max_features

Nice work

Stopword removal is a process used to eliminate common words that carry little semantic meaning from the text data. Removing these words can reduce the size of the data to be processed without losing significant information, thus improving efficiency.

5. How can you find the column index for a specific term in the document-term matrix using the vectorizer object? 1 / 1 point

- Using the transform method
- Using the vocabulary_ property
- Using the decode function
- Applying the fit function

Nice work

The vocabulary_ property of the vectorizer is a dictionary that maps terms to their respective column indices in the document-term matrix.

6. How would you find the term corresponding to a specific column in a document-term matrix using a vectorizer? 1 / 1 point

- Use get_feature_names
- Use fit_transform method
- Call toarray function

Nice work

The get_feature_names method of a vectorizer returns a list of all terms in the vocabulary, where the index in the list corresponds to the column number in the document term matrix.